

Compare and Order Integers

Lesson 9-3

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

-3 < 2 because -3 is farther left on the number line than 2

Compare

To see if a number is bigger, smaller, or equal to another.

Least to greatest: -5, -2, 0, 3, 7 (left to right on the number line)

Order

To put numbers in order, smallest to biggest or biggest to smallest.

4 > -1 because 4 is to the right of -1 on the number line

Greater than

Farther right on a number line. The > sign.

-6 < -2 because -6 is farther left — it is farther from zero

Less than

Farther left on a number line. The < sign.

-4 is 4 units to the LEFT of zero on the number line

Negative integer

A whole number smaller than zero, like -5.

..., -3, -2, -1, 0, 1, 2, 3, ...

Integer

Whole numbers and their opposites, like -2, -1, 0, 1, 2.

Key Ideas & Notes

- Captain Vega's crew found treasures at different depths underwater!
- The deeper the treasure, the more negative the elevation.
- A gold coin is at -2 meters, a silver crown is at -7 meters, and a ruby ring is at -5 meters.
- The crew needs to rank them from the shallowest (closest to the surface) to the deepest.
- Plot the treasure depths on the number line. Then order them from least to greatest.

Think About It

- Which treasure is closest to the surface?
- Is -7 greater than or less than -2?
- How does the number line help you compare negative numbers?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Why is -3 less than 2, even though 3 looks like a bigger number than 2?

Level 1 support

Start here: -3 is less than 2 because on the number line ____.

On the number line, -3 is to the ____ of 2.

En la recta numerica, -3 esta a la ____ de 2.

Numbers get smaller as you move ____.

Los numeros se hacen mas pequenos a medida que te mueves ____.

Word bank: integer compare number line less than greater than

Listen for: Student explains that -3 is farther left on the number line than 2, and numbers decrease moving left.

Level 2

Push students to explain why every negative number is less than every positive number.

- Every negative number is less than every positive number because ____.
- On the number line, all negatives are ____ of all positives.

EXPLORE

Order these temperatures from coldest to warmest: -8, 3, -1, 5. How does the number line help you put them in order?

Level 1 support

Start here: I can order the integers by looking at their position on ____.

The coldest temperature is ____ because it is farthest ____.

La temperatura mas fria es ____ porque esta mas a la ____.

From least to greatest, the order is ____.

De menor a mayor, el orden es ____.

Word bank: integer order compare number line least to greatest

Listen for: Student orders -8, -1, 3, 5 and uses left-to-right position on the number line to justify the order.

Level 2

Ask students to write a comparison statement using < or > for two of the temperatures and read it aloud.

- I can write ____ < ____ because ____.
- Reading left to right on the number line, ____ comes before ____.

CONNECT

How does knowing a number's position on the number line tell you whether to use the $<$ or $>$ symbol?

Level 1 support

Start here: The number farther to the left is always ____.

The number on the left is ____ than the number on the right.

El numero de la izquierda es ____ que el numero de la derecha.

The open side of the symbol points to ____.

El lado abierto del simbolo apunta a ____.

Word bank: compare order number line less than greater than

Listen for: Student connects left position to 'less than' and right position to 'greater than', and uses the symbol correctly.

Level 2

Push students to generalize a rule for comparing two negative numbers, like -7 and -2.

- When comparing two negatives, the one closer to zero is ____.
- So -7 is ____ than -2 because ____.

REFLECT

Two students disagree: one says -10 is greater than -2 because 10 is bigger. Who is right, and how do you know?

Level 1 support

Start here: The correct answer is ____ because ____.

-10 is actually ____ than -2.

-10 en realidad es ____ que -2.

I know because on the number line -10 is ____.

Lo se porque en la recta numerica -10 esta ____.

Word bank: integer compare number line less than greater than

Listen for: Student explains -10 is less than -2 because it is farther left, correcting the 'bigger digits' misconception.

Level 2

Ask students to critique why the 'bigger digits means bigger number' idea fails for negatives.

- The 'bigger digits' idea fails for negatives because ____.
- With negative numbers, more distance from zero actually means ____.

Guided Examples

Example 1

Which statement is TRUE?

Solution: -5 is farther to the left on the number line than -2, so $-5 < -2$ is true.

Answer: B. $-5 < -2$

Example 2

Which integer is the LEAST: -3, 5, -8, 2, 0?

Solution: -8 is farthest to the left on the number line, making it the least (smallest) integer.

Answer: A. -8

Example 3

What is $|-8|$?

Solution: Absolute value is distance from 0, always positive. $|-8| = 8$.

Answer: A. 8

Write About the Math

The Writing Revolution

I can explain my reasoning using the words compare, order, greater than, and less than.

1. Kernel Sentence subject + verb

Model: Compare is to see if a number is bigger, smaller, or equal to another.

Comparar es ver si un número es mayor, menor o igual que otro.

Write a kernel sentence about compare. Use a subject and a verb.

Escribe una oración base sobre comparar. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Compare matters in math

Comparar importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Compare matters in math because ____.

Comparar importa en matemáticas porque ____.

but
pero

Compare matters in math, but ____.

Comparar importa en matemáticas, pero ____.

so
entonces

Compare matters in math, so ____.

Comparar importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about compare.
Di un hecho verdadero sobre compare.

Compare ____.

Question *Pregunta*

Ask a question about compare.
Haz una pregunta sobre compare.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about compare.
Muestra entusiasmo sobre compare.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with compare.
Dile a un compañero qué hacer con compare.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

On the number line, -3 is to the ____ of 2.

En la recta numerica, -3 esta a la ____ de 2.

Numbers get smaller as you move ____.

Los numeros se hacen mas pequenos a medida que te mueves ____.

The coldest temperature is ____ because it is farthest ____.

La temperatura mas fria es ____ porque esta mas a la ____.

| Try It

Solve on your own. Check the answer key when you are done.

1. What is $|-8|$?

- A. 8
- B. -8
- C. 0
- D. -16

Show your work:

2. Three cities have temperatures of -12°F , -5°F , and 3°F . Which city is warmest?

- A. 3°F
- B. -5°F
- C. -12°F
- D. They are equal

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A student says: 'Negative numbers are always less than positive numbers, so -100 is less than 1.' Is this correct? Now the student says: '-100 is less than -1 because 100 is greater than 1.' Is THIS correct? Explain the difference.

Sentence starter: The first statement is ____ because _____. The second statement is ____ because with negative numbers, _____. On the number line, -100 is ____ of -1, so ____.

Show your work:

Reflect — Exit Ticket

Which list shows the integers in order from LEAST to GREATEST?

- A. -2, -5, 0, 3, 7
- B. 7, 3, 0, -2, -5
- C. -5, -2, 0, 3, 7
- D. 0, -2, -5, 3, 7

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 8 — *Absolute value is distance from 0, always positive. $|-8| = 8$.*
2. **Try It 2:** A. 3°F — *3°F is the greatest because it is farthest to the right on the number line. The order from coldest to warmest is -12, -5, 3.*
3. **Exit Ticket:** C. -5, -2, 0, 3, 7 — *-5 is the smallest (farthest left on the number line), then -2, then 0, then 3, then 7. The correct order from least to greatest is -5, -2, 0, 3, 7.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Compare is to see if a number is bigger, smaller, or equal to another.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).